MASTER SOFTWARE TEST PLAN

Manual & Automated Quality Assurance for SauceDemo (Swag Labs) E-Commerce Web Application

Cohort / Track	GIZ5_SWD6_S3 (DEPI Program - Software QA & Testing Track)
Target AUT	SauceDemo / Swag Labs (https://www.saucedemo.com/)
Test Methodology	Manual Functional, API, DB & Automated System Testing
Testing Team	Ahmed Reda (Lead), Amr Khaled, Mai Hussen, Hager Abdelrahem, Mohamed Mahmoud Taha, Kerolos Elia
Date of Execution	October 2026

1. Project Overview & Purpose

SauceDemo (Swag Labs) is an e-commerce platform utilized to demonstrate end-to-end user workflows, including authentication, product browsing, cart management, shipping detail collection, order verification, and session management. This Master Test Plan establishes the testing objectives, strategy, team responsibilities, execution scope, and exit criteria required to ensure full system stability and software quality.

Primary Objectives:

- • **Functional Accuracy:** Verify that all core user functionalities (login, product grid, cart, checkout, confirmation, logout) work according to defined acceptance criteria.
- • **Defect Discovery:** Identify, log, and track UI, functional, and session security defects in Jira.
- • **Multi-Tier Integrity:** Validate data integrity across frontend UI, Postman API collections, and MySQL database verification.
- • **Quality Gatekeeping:** Ensure release readiness through structured manual and automated regression testing.

2. Scope of Testing

2.1 In-Scope Features & Work Breakdown

Module & User Story	Assigned QA Engineer	Detailed In-Scope Verification Workflows

US-01: User Login	Ahmed Reda (Lead)	Validation of standard_user, locked_out_user, problem_user, performance_glitch_user; empty field validation, password masking, error banner assertions.
US-02: Browse Products	Amr Khaled	Product grid UI layout, price/image rendering, sorting options (A-Z, Z-A, Price Low-High, Price High-Low), product detail view navigation, cart icon counter.
US-03: Manage Shopping Cart	Ahmed Reda (Lead)	Adding/removing products from grid & product detail pages, dynamic button toggling ('Add to cart' / 'Remove'), cart list item display, empty cart state, 'Continue Shopping' navigation.
US-04: Checkout - Enter Information	Mai Hussen	Mandatory field checks (First Name, Last Name, Zip/Postal Code), inline validation error popups, field focus, and form cancellation routing.
US-05: Checkout - Order Overview	Hager Abdelrahem	Cart item list verification in overview, price calculation accuracy (Item Total sum, Tax rate computation, Grand Total verification), finish and cancel actions.
US-06: Order Confirmation	Mohamed Mahmoud Taha	Success checkmark display, 'Thank you for your order!' message verification, complete cart clear state, counter zeroing, redirection to home.
US-07: User Logout & Session	Kerolos Elia	Side hamburger menu operation, immediate session termination, redirection to login page, post-logout URL protection (preventing back-button access).

2.2 Out-of-Scope Features

- Real monetary transactions and live credit card authorization (simulated e-commerce environment).
- Native mobile application testing (focused on web application across desktop/responsive browsers).
- Load, stress, and volume testing (focused on manual, API, DB, and functional automation).

3. Testing Strategy & Methodology

3.1 Levels & Types of Testing

- • **Manual Functional Testing:** Documented test case execution across UI, forms, buttons, and user flows.

- • **Automation Testing:** Automating regression test execution using Selenium WebDriver, Java, TestNG, and Cucumber BDD framework.
- • **API Testing:** Validating backend endpoints, status codes (200, 400, 401), and payload responses using Postman.
- • **Database Testing:** Verifying backend data consistency (user records, product catalogs, order entries) using MySQL SQL queries.
- • **Regression Testing:** Executing end-to-end regression test suites after code changes and defect fixes.

3.2 Test Data Approach

- • **User Credentials:** Standard user: standard_user | Locked out user: locked_out_user | Password: secret_sauce.
- • **Catalog Items:** Sauce Labs Backpack ($29.99), Sauce Labs Bike Light ($9.99), Sauce Labs Bolt T-Shirt ($15.99), Fleece Jacket ($49.99), Onesie ($7.99), Red T-Shirt ($15.99).
- • **Form Inputs:** Valid text strings, special characters, numerical zip codes, and empty strings for negative testing.

4. Defect Management & Severity Matrix

All identified defects will be reported and tracked in Jira using the following severity classification:

Severity Level	Business & System Impact	SauceDemo Examples
Critical	Total blockage of core system functionality; system crash or page unaccessibility. No workaround available.	Login page completely fails to load (500 Error); 'Add to Cart' button unclickable; Checkout process crashes.
High	Major feature defect or incorrect business logic calculation. System is usable but critical path is broken.	Wrong Grand Total calculation (Item Total + Tax mismatch); Order confirmation email not triggered; Broken product image links.
Medium	Minor feature dysfunction, layout overflow, or pagination issue. System usability is preserved.	Button misalignment on Product page; Product description text slightly overlapping; Minor sort ordering anomaly.
Low	Cosmetic, typo, or minor design deviation with negligible business impact.	Spelling typo in product description; Minor color shade mismatch from design mockup; Missing favicon.

5. Test Environment, Tools & Timeline

5.1 Technical Infrastructure & Toolstack

Application Under Test (AUT)	SauceDemo / Swag Labs (https://www.saucedemo.com/)
Operating Systems & Browsers	Windows 11 / Windows 10 | Google Chrome (Latest), Mozilla Firefox, MS Edge
Automation Framework	Java JDK 17, Selenium WebDriver, TestNG, Cucumber BDD, IntelliJ IDEA
API & Database Tools	Postman (API Collections & Assertions), MySQL 8.0 (Database Verification)
Test Management & Version Control	Jira (Defect & Sprint Tracking), GitHub (Repository & CI Integration)
Reporting Tools	Allure Reports, Extent Reports, MS Word & Excel (Test Execution Sheets)

5.2 Timeline & Execution Milestones

- • **Week 1:** Test Planning, Requirement Analysis, Architecture Breakdown, and Environment Configuration.
- • **Week 2 - 3:** Manual Test Execution (US-01 to US-07), Test Case Documentation, and Initial Jira Defect Logging.
- • **Week 4:** API Testing with Postman, Database SQL Script Execution, and Selenium Test Suite Development.
- • **Week 5:** Automated Regression Execution, Defect Retesting, and User Acceptance Testing (UAT) Support.
- • **Final Week:** Test Summary Report Generation, Quality Sign-Off, and Final Deliverables Submission.

6. Entry & Exit Criteria

6.1 Entry Criteria

- SauceDemo web application URL is accessible and stable.
- Test Plan document and User Story Acceptance Criteria are reviewed and approved.
- Test environment, browsers, Postman collections, and Selenium setups are fully operational.
- Test data (user accounts, catalog items, shipping addresses) is prepared.

6.2 Exit Criteria

- 100% of planned test scripts (Manual, API, Automated) are executed.
- Minimum 90% overall Pass Rate across all executed test suites.

- Zero open Critical or High severity defects remaining in Jira.
- All Medium and Low defects are documented, categorized, and deferred to future releases.
- Final Test Execution Summary Report compiled and approved by QA Lead.

7. Risk Management & Dependencies

- • **Risk: Target Environment Downtime -> Mitigation:** Maintain fallback offline/local test environments and local mock setups.
- • **Risk: Unresolved Critical Bugs -> Mitigation:** Daily sync meetings with QA Lead to prioritize bug fixes and conduct rapid retests.
- • **Risk: Script Flakiness -> Mitigation:** Automated test scripts maintained in GitHub repository with modular page-object design.

8. Test Deliverables & Approvals

8.1 List of Deliverables

- **1. Test Plan:** Master Test Plan Document (This Word Deliverable).
- **2. Test Cases Specification:** Detailed Manual & BDD Test Scenarios and Test Cases.
- **3. API & DB Scripts:** Postman Collection Files (.json) & SQL Script Files (.sql).
- **4. Automation Suite:** Selenium WebDriver + TestNG Automation Project Repository.
- **5. Defect Reports:** Jira Bug Reports with Steps to Reproduce and Screenshots.
- **6. Test Summary Report:** Final Metrics, Pass/Fail Breakdown, and Quality Sign-off.

8.2 Sign-off & Approval

Role	Name	Sign-Off Status / Date
QA Lead / Lead Author	Ahmed Reda	APPROVED - October 2026
QA Review Team	Amr Khaled, Mai Hussen, Hager Abdelrahem, Mohamed Mahmoud Taha, Kerolos Elia	APPROVED - October 2026